

General Features

Sensor	3-component orthogonal 5Hz high sensitivity geophone (10Hz optional)
Digitalization	3x 24-bit $\Delta\Sigma$ ADC with GNSS disciplined clock
Storage Capacity	32G (2ms sampling, 45days continuous recording) 64G optional
Node Autonomy	Internal 69WH/80WH lithium-ion battery pack supports 28/33 days continuous records
Timing Accuracy	$\pm 10\mu s$ (continuous GNSS disciplined) $\pm 1ms$ (within 1 hour without GNSS signal) (External active GNSS antenna optional)
Working Mode	Autonomy with wireless on-site QC inspection, Support UAV patrol
LED Indicator	Nodal health, GNSS synchronization status, Wireless data harvesting status
Automated Tests	Internal testing signal generator for function and performance self-inspection and diagnosis
Data Harvest	Infield: Wireless/Portable download cable Indoor : Data download/battery charging unit
Infield Management	Handheld Android tablet
Weight	1.8 kg incl. internal battery and 3-C geophones
Size	Diameter: 12cm Height: 16cm
Operating Temperature	$-40^{\circ}C \sim +70^{\circ}C$
Waterproof	IP68

Acquisition Specifications

ADC Resolution	24-bit
Sample Interval	0.5ms、1ms、2ms、4ms、10ms
Preamplifier Gain	x1、x2、x4 (0dB、6dB、12dB)
Gain Accuracy	0.25%
Full Scale Input Signal	$\pm 2.5V_{p-p}$ @ 0dB
Instantaneous DR @2ms(typical value)	125dB @ 0dB
Equivalent Input Noise @2ms(typical value)	$1\mu V_{rms}$ @ 0dB
Harmonic Distortion	-120dB @ 31.5Hz
Frequency Response	0Hz ~ 866Hz @ 0.5ms (-3dB) 0Hz ~ 800Hz @ 0.5ms (-0.1dB)
Anti-alias Filter	Linear filter, 86.6% of Nyquist -3dB bandwidth
Stopband Attenuation	>105dB @Nyquist
Common Mode Rejection	>95dB



Product Highlights

- The most compact 3C node on the market with internal orthogonal high sensitivity geophones (5Hz or 10Hz);
- Three channels of 24-bit $\Delta\Sigma$ ADC with configurable preamplifier gain and sampling rate;
- High sensitive GNSS module equipped with active patch antenna supports GPS, Beidou, and GLONASS;
- Built-in test signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- Built-in BLE module together with LEDs for onsite configuration and proximity, supports vehicle (20 meters) or UAV(100 meters) patrol;
- Industrial Android handheld tablet supports working status verification after deployment, including battery usage, environmental temperature, tilting, GNSS status, real-time seismic waveform, etc.;
- P&R (Place & Record) mode for rapid and convenient field deployment, supporting automatic GNSS position matching with line and point number;
- High strength engineering plastic enclosure for enhanced anti-dropping capability, waterproof (IP68), flame retardant (5VA), ultraviolet aging (F1).



High Sensitivity Geophone

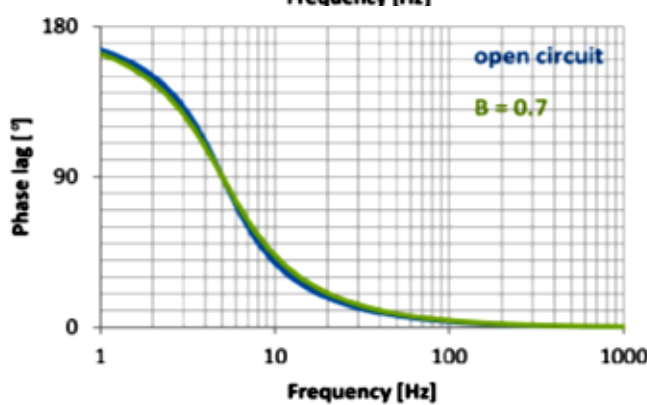
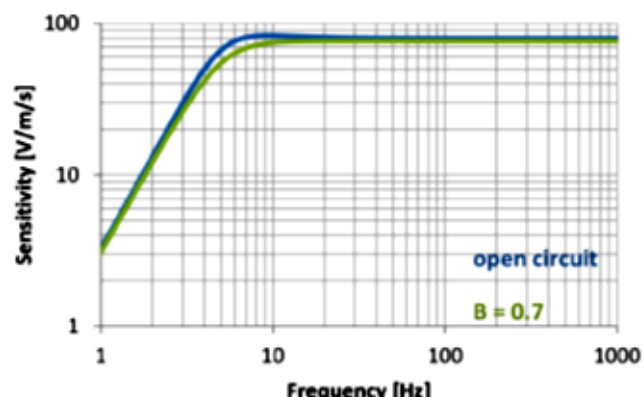
5Hz high sensitivity geophone (SM5)

Natural Frequency	5Hz±7.5%
Sensitivity	80V/m/s±7.5%@0.6 damping 76.5V/m/s±7.5%@0.7 damping
Open-circuit damping	0.6
Spurious	>150Hz
Harmonic Distortion	0.1%@0.7 V/in/s p-p (12Hz)
Coil resistance	1850Ω±5%

10Hz high sensitivity geophone (SM24HS)

Natural Frequency	10Hz±3.5%
Sensitivity	85.8V/m/s±5%@0.475 damping 78.8V/m/s±5%@0.7 damping
Open-circuit damping	0.475
Spurious	>240Hz
Harmonic Distortion	0.1%@0.7 V/in/s p-p (12Hz)
Coil resistance	1800Ω±5%

*Unless otherwise stated, the above parameters are measured at 22°C(71.6°F) ambient temperature in the vertical/horizontal attitude of the geophone.



5Hz High Sensitivity Geophone Response Curve and Phase Lag

Stackable Charging Unit

Slots	12
Charging Time	4 hours
Charging Power	350W
External Dimensions (W x H x D)	48.3cm x 4.45cm x 45.5cm
Operating Temperature	0°C ~ +40°C



Stacked Data Download Unit (1U)



Stacked Charging Unit (1U)

Stackable Data Download Unit

Slots	9
Slot Download Rate	600MB/min.
Full Download Rate	5.4GB/min.
External Dimensions (W x H x D)	48.3cm x 4.45cm x 45.5cm
Operating Temperature	0°C ~ +40°C

*The width and thickness of the stackable charging unit and the data download unit meet the installation requirements of standard 19-inch rack.



Switch

Optional Leveling Tripod

Quadri-Spike